DIVIT RAWAL

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EDUCATION

University of California, Berkeley 2023 – 2027

B.A. Statistics, Computer Science, Physics *Berkeley, CA*

Graduate-Level Coursework: Statistical Learning Theory; Theoretical Statistics; Mathematical Statistics; Randomness and Computation; Learning for Dynamics and Control; Randomized Linear Algebra, Optimization, Large-Scale Learning.

Teaching: EE 122 (Communication Networks) Head TA (Sp26), Reader (Sp25). Developed lab curriculum, wrote and graded exams and homeworks, and led weekly recitation sections.

PUBLICATIONS AND PREPRINTS

[1] **Majority-of-Three is Optimal**
Divit Rawal, Nikita Zhivotovskiy.
Under review, Transactions on Machine Learning Research (TMLR). [arXiv]

[2] **A Theory of Saddle Escape in Deep Nonlinear Networks**
Divit Rawal, Michael R. DeWeese.
Under review, Journal of Machine Learning Research (JMLR). [arXiv]

[3] **Rao-Blackwellized Score Matching on Manifolds**
Divit Rawal.
International Conference on Machine Learning (ICML) 2026, Workshop on Structured Probabilistic Inference and Generative Modeling. [arXiv]

[4] **ALPHANSO: Open-Source Modeling of (α,n) Neutron Source Terms**
Divit Rawal, Anthony J. Nelson, William Zywiec, Daniel Siefman.
Nuclear Instruments and Methods in Physics Research, Section A. [doi]
Appeared at ANS Student Conference 2026 and INMM Annual Meeting 2026.

AWARDS AND GRANTS

Best Paper Award, American Nuclear Society Student Conference Apr 2026

Best Paper in Mathematics, Computation, and AI Applications for "ALPHANSO: Open-Source Modeling of (α, n) Source Terms."

VESSL AI Academia Grant Aug 2025

For independent research into geometric deep learning and information geometry; sole undergraduate recipient among cohort of Ph.D. students and postdoctoral researchers. (Supported work on [3].)

RESEARCH EXPERIENCE

Berkeley Statistical Machine Learning Group Jun 2026 – Present

Researcher (advisor: Nikita Zhivotovskiy) *Berkeley, CA*

Resolved conjecture proving that majority voting over three consistent classifiers achieves optimal PAC learning rates in the realizable setting; authored resulting paper [1].

Currently working on sample compression schemes and efficient optimal online learning algorithms.

Berkeley Artificial Intelligence Research (BAIR) Sep 2024 – Present

Researcher (advisor: Michael DeWeese) *Berkeley, CA*

Empirically observed existence of in-context learning behavior in kernel ridge regression and analyzed emergence via spectral and random matrix theory methods.

Derived layer-imbalance identity for nonlinear networks, extending deep-linear network analysis and predicting first feature acquisition time in small-initialization neural networks; authored resulting paper [2].

Lawrence Livermore National Laboratory Mar 2025 – Present

Research Scientist, Nuclear Science and Security Consortium Affiliate (Oct 2026–Present) *Livermore, CA*
Nuclear Science and Security Consortium Fellow (Mar 2025–Sep 2026) (advisor: Daniel Siefman)

Developed ALPHANSO [4], an open-source framework for deterministic modeling of (α, n) neutron production using continuous slowing-down approximations and modern evaluated nuclear data.

Developing new capabilities for FREYA event-by-event fission model, improving simulation of correlated neutron and gamma emission for radiation transport applications.

Invited Talks

ALPHANSO: Open-Source (α,n) Neutron Source Terms

Lawrence Livermore National Laboratory	Mar 2026
Lawrence Berkeley National Laboratory	Apr 2026
University of Tennessee, Knoxville	Jun 2026

Industry Experience

Cisco Systems, Foundation AI Jun – Aug 2025
Software Engineer Intern *San Francisco, CA*

Developed components of the post-training and evaluation pipeline for cybersecurity-specialized LLMs based on Llama 3.1 8B, including instruction tuning and RLHF.

Contributed to the open-weight release of Foundation-Sec-8B and Foundation-Sec-8B-Instruct, including technical reports on data curation, training methodology, and benchmark evaluation.

Amazon Aug – Dec 2023
Software Engineer Intern *Remote*

Diagnosed and resolved production pipeline failures affecting 1M+ users; implemented a k-means clustering module in Java for OpenSearch ml-commons.

Technical Skills

Programming: Proficient in Python, C, C++, Java, SQL; intermediate in MATLAB, FORTRAN, Julia, R.

ML/AI: Proficient in PyTorch; intermediate in JAX.

Other: Intermediate proficiency in Lean, Mathematica.